Understanding duplicates Values

by Shahriar Analytical Academy

What Are Duplicate Values in Data?

Definition

Duplicate values in a dataset are **identical records** or rows that appear more than once. These may occur across **entire rows** or within **specific columns**.

Two Types of Duplicates

1. Full Duplicates: Every value in the row is exactly the same as another row.
2. Partial Duplicates: Certain key columns (like ID, name, email) are repeated, but other fields may differ.

Real-World Example:

Imagine a customer database where the same customer was entered twice:

Row 1: John Doe, johndoe@email.com, 123 Main St
Row 2: John Doe, johndoe@email.com, 123 Main St

These are full duplicates.

Why Duplicate Values Occur

1 **Manual Data Entry**

Repeated entry of the same information by different users or at different times.

2 **Data Collection Errors**

APIs or web scrapers may fetch the same record multiple times due to server bugs or pagination issues.

3 **Merging Datasets**

Combining data from multiple sources or systems without proper de-duplication checks.

4 **System Glitches or Sync Failures**

Some systems might save multiple copies of records during syncing or backup failures.

Why Duplicates Are a Problem

Skewed Analysis

Repeated values inflate totals, averages, and counts. For example, the same sales transaction counted twice increases total sales incorrectly.

Incorrect Insights

Duplicate customer records might lead to sending multiple emails to the same person, creating a poor user experience.

Poor Data Quality

Machine learning models and statistical methods depend on clean data. Duplicates introduce bias and noise.

Increased Storage and Cost

More rows mean more storage, especially in large enterprise databases — leading to slower performance and higher costs.

Summary – Key Learning Points

1

Understanding Duplicates

Duplicate values occur when identical or nearly identical entries appear multiple times in a dataset.

2

Common Causes

They arise due to manual input, system errors, or merging datasets.

3

Impact on Analysis

Duplicates distort analysis, reports, and modeling efforts.

4

Detection Methods

Detection tools include Excel, Python (duplicated()), and grouping logic.

5

Careful Removal

Removal must be done carefully — some "duplicates" might actually be valid records.

6

Prevention Strategies

Preventive strategies include validation rules, unique constraints, and regular audits.

Thank you!

Shahriar Analytical Academy